

JEE MAIN REVISION MASTER: MATRICES AND DETERMINANTS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 65 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Properties of Determinants & System of Linear Equations	38 PYQs	Determinants & Linear Systems
Very High	Matrix Inversion, Adjoint & Powers	29 PYQs	Matrix Algebra

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Properties of Determinants & System of Linear Equations	$\det(kA) = k^n \det(A)$, $\text{Unique: } \Delta \neq 0$, $\text{Infinite: } \Delta = \Delta_x = \Delta_y = \Delta_z = 0$, $\text{No sol: } \Delta = 0 \text{ \& at least one } \Delta_i \neq 0$
Matrix Inversion, Adjoint & Powers	$A \cdot \text{adj}(A) = (\det A) I_n$, $\det(\text{adj}(A)) = (\det A)^{n-1}$, $\text{adj}(\text{adj}(A)) = (\det A)^{n-2} A$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Properties of Determinants & System of Linear Equations (Very High Yield — 38 PYQs)

Core Principle: Evaluation of determinants, Cramer's rule, and consistency conditions for non-homogeneous and homogeneous linear systems.

■■ **Caution / JEE Pitfall:** For an $n \times n$ matrix A , $\det(kA) = k^n \det(A)$, NOT $k \det(A)$!

■ **Speed Trick:** A homogeneous system $AX = 0$ has non-trivial (infinite) solutions if and only if $\det(A) = 0$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The area of the triangle with vertices $(3,8)$, $(-4,2)$ and $(5,1)$ is $(P/4)$, then the value of P is..." [Ans: C]
- [2026 Shift] Q#1 (MCQ): "Consider the following statements: Statement I: If A is a non-singular matrix, then A^{-1} exists. Statement II: If A and B are symmetric ..." [Ans: N/A]
- [2026 Shift] Q#1 (MCQ): "Which of the following is correct?..." [Ans: C]

3.2 Matrix Inversion, Adjoint & Powers (Very High Yield — 29 PYQs)

Core Principle: Properties of adjoint and inverse matrices, matrix multiplication transpose $(AB)^T = B^T A^T$, and symmetric/skew-symmetric decomposition.

■ **Caution / JEE Pitfall:** A skew-symmetric matrix of ODD order has determinant equal to ZERO ($\det(A) = 0$).

■ **Speed Trick:** For a 3×3 matrix: $\det(\text{adj}(A)) = (\det A)^2$, and $\det(\text{adj}(\text{adj}(A))) = (\det A)^4$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Consider the following statements: Statement I: If A is a non-singular matrix, then A^{-1} exists. Statement II: If A and B are symmetric ..." **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "Match List-I with List-II List-I List-II a) A matrix which is not a square matrix i) Symmetric matrix b) A square matrix $A=A^T$ ii) Null ..." **[Ans: N/A]**
- **[2026 Shift] Q#3 (MCQ):** "Let X be a matrix of order $2 \times n$ and Z be a matrix of order $2 \times p$. If $n=p$, then the order of the matrix $8X-9Z$ is:..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The area of the triangle with vertices $(3,8)$, $(-4,2)$ and $(5,1)$ is $(P/4)$, then the value of P is

- (A) $(61/2)$
- (B) $(2/61)$
- (C) 122 **[Correct]**
- (D) $(1/122)$

Correct Answer: C

Solution Analysis: Using the area formula for a triangle with vertices (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , $\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$. Here the vertices are $(3,8)$, $(-4,2)$ and $(5,1)$. So, $\text{Area} = \frac{1}{2} |3(2-1) + (-4)(1-8) + 5(8-2)| = \frac{1}{2} |3(1) + (-4)(-7) + 5(6)| = \frac{1}{2} |3 + 28 + 30| = \frac{1}{2} \times 61 = \frac{61}{2}$. Given that the area is $(P/4)$, $(P/4) = \frac{61}{2}$. Multiplying both sides by 4, $P = 122$. So, the correct option is 122 Option C.